

Non-Euclidean Normative Space: When Legal Hierarchies Curve

Extended Phenotype Theory, Lateral Silencing, and the Geometry of Legal Systems Beyond Kelsen

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Abstract

Legal positivism, from Kelsen through Hart and their contemporary successors, operates on an implicit geometric assumption: the normative space in which law functions is flat.

The Kelsenian pyramid presupposes five properties of Euclidean normative space: each norm has a single declared function; validity flows in one direction, from superior to inferior; the causal effects of a norm are proportional to its hierarchical rank; formal invalidity tracks functional failure; and norms in distinct regulatory domains operate on parallel tracks that never intersect.

This paper argues that these properties hold only in a limiting case, specifically, in systems where norms are structurally enforceable, where enforcement is systematic rather than discretionary, and where the secondary effects of norms are negligible relative to their primary regulatory functions. When those conditions are not satisfied, the normative space curves.

Also, this paper develops a framework for analyzing the geometry of curved normative spaces, drawing on three theoretical resources that have not previously been combined in this way: the Extended Phenotype Theory of Law (EPT; Lerer, 2026a), which identifies legal norms as cultural replicators whose effects extend well beyond their declared regulatory domain; the Lateral Silencing Exaptation (LSE; Lerer, 2026b), which formalizes the mechanism by which structurally unenforceable norms suppress political contestation across domains; and the mathematical tradition of non-Euclidean geometry, understood not as a technical apparatus but as a conceptual framework for thinking about spaces in which the standard axioms fail.

The paper proceeds in five stages. First, it reconstructs the implicit Euclidean geometry of the dominant traditions in legal theory, showing that Kelsen, Hart, Fuller, Raz, and their contemporary successors all presuppose flat normative space, even when they disagree on every substantive question. Second, it identifies the conditions under which normative space curves, developing a taxonomy of curvature types corresponding to the three failure modes identified in LSE theory. Third, it proposes three specific non-Euclidean geometries as candidates for modeling curved normative spaces: Riemannian geometry for systems of high lateral silencing density; hyperbolic geometry for systems of high modularity and domain separation; and a quantum superposition model for norms whose functional state is indeterminate until the enforcement act collapses it. Fourth, it connects this geometric framework to the existing diagnostic instruments of the EPT research program, showing how the Constitutional Lock-in Index, the Institutional Hysteresis Rate, and the Lateral Silencing Index can be understood as measures of curvature in normative space. Fifth, it considers the implications of non-Euclidean normative space for legal theory, legal design, and the philosophy of law.

The paper closes with five falsifiable predictions.

Keywords: non-Euclidean geometry, normative space, Kelsen, Hart, Fuller, extended phenotype, lateral silencing exaptation, constitutional lock-in, legal theory, philosophy of law, evolutionary jurisprudence

I. Introduction: The Geometry Nobody Examined

In 1854, Bernhard Riemann delivered his *Habilitationsvortrag* at Göttingen, a lecture on the foundations of geometry that most of his audience found incomprehensible. The problem he addressed had seemed, until then, not to be a problem at all. Euclid's geometry had worked so well for so long, and matched spatial intuition so perfectly, that the question of whether it was the only possible geometry had not seriously been asked. Riemann showed not only that other geometries were possible but that the geometry of the space in which physical objects actually exist was an empirical question, not a logical necessity. Euclidean geometry was a special case, valid in flat spaces, and the universe turned out to inhabit a curved one.

Legal theory has not had its Riemann. The question of whether normative space has a geometry, and what kind of geometry it might be, has not been systematically asked. This is not because legal theorists have been incurious about the structure of law. Kelsen's hierarchy, Hart's union of primary and secondary rules, Fuller's eight principles of legality, Dworkin's rights as trumps, Raz's service conception of authority: these are all, in different ways, accounts of the structure of normative systems. But they all share, without examination, the assumption that the space in which norms operate is flat. That the effects of norms flow in straight lines. That hierarchies are genuinely pyramidal. That norms in different regulatory domains run on parallel tracks.

This paper argues that those assumptions hold only in a limiting case, and that a large and practically important class of normative systems operates in curved spaces where the standard axioms fail. The curvature is not a metaphor or a loose analogy. It corresponds to specific, identifiable features of regulatory environments that can be measured with existing instruments and predicted in advance. When normative space curves, the standard tools of legal analysis, designed for flat spaces, produce systematically distorted conclusions. Understanding the geometry is not an academic luxury. It is a precondition for accurate diagnosis.

The argument proceeds from a concrete phenomenon identified in prior work in this research program. The Lateral Silencing Exaptation (LSE; Lerer, 2026b) describes a class of norms whose primary regulatory function has failed but which persist because they generate a secondary function: the suppression of political contestation by regulated agents who are permanently in technical violation. This secondary function operates laterally, across regulatory domains that are formally unconnected. A structurally unenforceable environmental norm silences firms not only

on environmental policy but on labor regulation, trade policy, and fiscal matters. The silencing travels across the putative parallels of the normative space. In Euclidean geometry, parallel lines do not meet. In the normative space created by LSE, they do.

That observation is the entry point for a more general inquiry. If norms can produce effects that travel laterally across domains, the flat geometry of standard legal theory is wrong, not as a matter of philosophical taste but as a matter of descriptive accuracy. The question then becomes: what geometry describes normative spaces that exhibit this kind of curvature? And what are the implications for how we understand legal validity, institutional design, and the relationship between formal legal structure and actual regulatory behavior?

The paper proceeds as follows. Section II reconstructs the implicit Euclidean assumptions of the dominant traditions in legal theory, working through Kelsen, Hart, Fuller, Raz, and their contemporary successors. Section III introduces the concept of normative curvature, identifying the conditions under which it arises and developing a taxonomy of curvature types. Section IV develops three specific non-Euclidean geometries as candidates for modeling curved normative spaces, connecting each to identifiable features of real regulatory systems. Section V connects the geometric framework to the existing diagnostic instruments of the EPT research program. Section VI develops the theoretical implications for legal philosophy, institutional design, and empirical legal studies. Section VII states the falsifiable predictions. Section VIII concludes.

II. The Euclidean Geometry of Standard Legal Theory

II.1 Kelsen: The Pure Pyramid

Hans Kelsen's Pure Theory of Law is the most geometrically explicit account of normative structure in the legal theoretical canon. The *Stufenbau*, the stepped construction of the legal order, is an explicitly hierarchical and unidirectional structure. Each norm derives its validity from the norm immediately above it. The chain of validity ascends through the hierarchy until it reaches the *Grundnorm*, the basic norm that is presupposed rather than derived. The *Grundnorm* is not itself a positive norm; it is the transcendental condition of the possibility of the legal order as a unity.

The geometric properties of this account are precise. The normative space is stratified: norms occupy levels, and the levels form a strict partial order. Validity flows downward, from higher to lower norms, and never in any other direction. A norm at level n cannot confer validity on a norm at level $n+1$, and it cannot affect, in any normatively relevant way, norms at the same level in a different regulatory domain. The geometry is pyramidal, with a single apex, planar levels, and no lateral connections between domains.

Kelsen is explicit that the question of the efficacy of a norm, whether it is actually followed or enforced, is distinct from the question of its validity. A norm can be valid even if it is never obeyed, and it loses validity only when its inefficacy becomes general and persistent enough that

the legal system as a whole can no longer be regarded as efficacious. This separation of validity from efficacy is a central feature of the pure theory, and it is also a central source of its geometric limitations. If efficacy is irrelevant to validity, then the actual effects of a norm in the social world, including effects on the behavior of regulated agents in domains formally unrelated to the norm's subject matter, cannot appear in the theory's account of normative structure. They are, by construction, outside the picture.

This is not a criticism of Kelsen's project, which was to give an account of legal validity, not of social causation. But it means that the Kelsenian framework is constitutively unable to represent the kind of normative curvature that the LSE mechanism generates. A norm that is formally valid but substantively unenforceable, and that produces political silencing as a secondary effect, is, within the Kelsenian framework, simply a valid norm that happens to have low efficacy. The secondary function is invisible to the theory. The curvature is assumed away by the initial abstraction.

The paper on the Grundnorm as evolutionary attractor in this research program (Lerer, 2026c; SSRN 5696484) showed that the Grundnorm can be reinterpreted as a Nash equilibrium in the evolutionary game of norm adoption, a fixed point around which institutional behavior stabilizes. That reinterpretation preserves the Kelsenian structure while giving it evolutionary content. The present paper goes further: it argues that the curvature of normative space cannot be accommodated within the Kelsenian geometry even with that reinterpretation, because the curvature arises from features of normative systems, their lateral silencing effects, their enforcement asymmetries, their secondary functions, that the Kelsenian abstraction by design excludes.

II.2 Hart: Rules About Rules in Flat Space

H.L.A. Hart's account of law in *The Concept of Law* (1961) is, in important respects, a critique of the Kelsenian picture. Hart rejects the command theory of law associated with Austin, and he finds Kelsen's Grundnorm philosophically unsatisfactory. His alternative centers on the distinction between primary rules, which impose duties and confer powers on ordinary subjects, and secondary rules, which are rules about rules: the rule of recognition, which identifies which norms are valid; the rule of change, which specifies how norms may be created and modified; and the rule of adjudication, which specifies how norm violations are to be determined and remedied.

Hart's account is more sociologically grounded than Kelsen's. The rule of recognition is not a transcendental presupposition but a social fact: it exists when officials of the legal system accept it as a standard for their own conduct and use it to identify valid law. This social grounding makes Hart's account more empirically tractable than Kelsen's. But it does not change the fundamental geometry. The normative space in Hart's account is still flat. Secondary rules operate at a higher level than primary rules, in the sense that they are about primary rules, but the space remains stratified and the connections between domains remain formally parallel.

Hart's texture of open texture argument, the observation that legal rules have a penumbra of cases in which their application is uncertain, introduces some flexibility into the picture. In hard cases, courts must exercise discretion, and that discretion can connect previously separate domains. But this is a feature of interpretation within the flat space, not a property of the space itself. The geometry remains Euclidean; the question is only about where exactly the boundaries of each norm fall within it.

Within the EPT research program, Hart's secondary rules have been interpreted as extended phenotypes of the primary-rule replicators: the rule of recognition is the institutional structure that allows replicators to identify which norms belong to their lineage; the rule of change is the mechanism by which replicators generate variants; the rule of adjudication is the selection mechanism that determines which variants persist. This interpretation is developed in the program's foundational paper (Lerer, 2026a; DOI 10.5281/zenodo.18870552). The geometric point being developed here is complementary: within that EPT interpretation, the secondary rules function as instruments for measuring the curvature of normative space, but they were designed for flat spaces and produce systematically distorted readings when the space curves.

II.3 Fuller: Legality as Flat Space Conditions

Lon Fuller's account of the internal morality of law in *The Morality of Law* (1964) is, in geometric terms, the most precise statement of the conditions under which normative space is flat. Fuller identifies eight principles of legality that any system must satisfy to count as law at all. These include the requirement that rules be promulgated, that they not be retroactive, that they be understandable, that they not be contradictory, that they not require the impossible, that they be stable enough to be relied upon, and that there be congruence between official action and declared rules.

The eighth principle, that rules not require the impossible, is the most directly relevant to the present argument. Fuller's account treats the impossibility of compliance as a defect of legality, a failure of the system to satisfy the conditions of genuine normative governance. A rule that cannot be obeyed does not guide conduct; it cannot be the kind of standard that Fuller regards as constitutive of law. On this account, structural unenforceability is a pathology that falls outside the category of law properly understood.

This is a principled position, and it is not wrong on its own terms. But it is descriptively incomplete in a way that the LSE framework reveals. Fuller's account assumes that the failure of the impossibility condition is a failure of the legal system considered in isolation, a defect in the rule's capacity to guide conduct. It does not consider the possibility that structural unenforceability might be a functional feature of the normative environment, selected and maintained precisely because it produces effects, specifically, political silencing, that a complied-with rule would not produce. Fuller sees the violation of his eighth principle as the normative space collapsing into

incoherence. The LSE framework shows that the space does not collapse: it curves. The curvature is the mechanism by which the violation is sustained.

More precisely: Fuller's eight principles are the conditions under which normative space is flat. When the eighth principle is violated, the space is curved. The other seven principles describe the remaining structure of a curved space: promulgation, non-retroactivity, understandability, non-contradiction, stability, and official congruence can all be satisfied in a curved normative space. What changes is the geometry of the effects. Fuller saw the violation of his conditions as a failure of law. The present framework treats it as a transformation of the geometric structure in which law operates.

II.4 Raz, Dworkin, and Contemporary Positivism: Flat Space Refined

Joseph Raz's service conception of authority, developed in *The Morality of Freedom* (1986) and subsequent work, offers a sophisticated account of when legal authority is legitimate and what it requires. On Raz's account, authority is justified when following the authority's directives enables subjects to better conform to the reasons that already apply to them than they could by reasoning directly from those reasons. This is the normal justification thesis. The pre-emption thesis follows: authoritative directives replace, rather than add to, the reasons that apply to subjects.

Raz's account is not explicitly geometric, but it has strong geometric implications. If legal directives pre-empt the underlying reasons that justify them, then the normative space is stratified in a specific way: the legal level sits above the level of first-order reasons, and its relationship to that level is one of pre-emption rather than mere coordination. The parallel tracks of different regulatory domains remain distinct: a directive about environmental compliance pre-empts the underlying environmental reasons; a directive about labor relations pre-empts the underlying labor reasons; there is no mechanism in Raz's account by which one domain's directive affects the pre-emptive force of another's.

Ronald Dworkin's interpretive account of law, developed in *Law's Empire* (1986), rejects the positivist separation between law and morality and argues that legal interpretation is always an exercise in constructive interpretation, presenting the law in its best light. The geometry of Dworkin's account is less explicitly hierarchical than Kelsen's or Hart's, but it remains flat in the relevant sense: the interpretive process operates within a domain, connecting norms in the same domain through principles that best justify the legal practices as a whole. Lateral effects across domains are not part of the framework.

Contemporary legal positivists, including Coleman, Shapiro, and Himma, have developed increasingly sophisticated accounts of the social foundations of law, the nature of legal systems, and the relationship between law and morality. These accounts differ substantially from each other on substantive questions. But they share, with Kelsen, Hart, Fuller, and Raz, the assumption of flat normative space. The connections they explore are vertical, hierarchical, or within-domain. The

lateral connections that characterize curved normative spaces are not part of any of these frameworks.

II.5 The Five Euclidean Axioms of Standard Legal Theory

The preceding survey allows the Euclidean assumptions of standard legal theory to be stated precisely. They are five in number, corresponding to the five properties of flat normative space.

Axiom 1 (Functional uniqueness): Each norm has a single declared function, which is the function it was enacted to serve. The norm's normative meaning is exhausted by that function. Secondary effects, if they exist, are causally relevant but normatively invisible.

Axiom 2 (Unidirectional validity): Validity flows in one direction, from higher to lower norms in the hierarchy. A lower norm cannot affect the validity of a higher norm. Causal influence in the reverse direction, from lower to higher, is possible as a sociological matter but not as a normative one.

Axiom 3 (Proportional effect): The normative effects of a norm are proportional to its position in the hierarchy. Constitutional norms have more normative force than statutory norms, which have more than regulatory norms. Effects do not amplify or distort as a function of enforcement structure.

Axiom 4 (Validity-efficacy independence): A norm's validity is independent of its efficacy. A structurally unenforceable norm that is never obeyed is still valid until formally repealed. Its failure to guide conduct is a sociological fact without normative consequences, except in the limiting case where the system as a whole loses efficacy.

Axiom 5 (Domain parallelism): Norms in distinct regulatory domains run on parallel tracks. An environmental norm does not affect the normative force of a labor norm. Cross-domain effects are possible as causal matters but not as normative ones. There is no mechanism in the structure of law by which one domain's norm curves the normative space of another.

These five axioms are the legal-theoretic equivalents of Euclid's postulates. They are intuitive, internally consistent, and adequate for the analysis of simple, well-functioning legal systems. They are the geometry of flat normative space. The question this paper addresses is what happens when any of them fails.

III. Normative Curvature: Conditions and Taxonomy

III.1 What Curves Normative Space

Normative space curves when the secondary effects of norms become normatively significant, specifically when those effects are large enough to alter the behavior of regulated

agents in ways that feed back into the normative structure itself. Three conditions, individually sufficient and jointly amplifying, produce curvature.

The first condition is structural unenforceability. When a norm requires conduct that is technically or economically impossible for the regulated population, it places all regulated agents in permanent technical violation. This creates a standing enforcement discretion that has no equivalent in systems of enforceable norms. In an enforceable system, the regulator's enforcement choices are constrained by the actual distribution of violations: enforcement can be targeted at actual violators, and a compliant actor has no enforcement exposure. In a structurally unenforceable system, the regulator can target any actor at any time, because all are in violation. The normative space is no longer flat in the enforcement dimension.

The second condition is discretionary enforcement. Structural unenforceability produces curvature only if enforcement is selective rather than universal. If all violations were prosecuted, the universal violation would produce universal collapse of the system rather than curvature. Curvature requires that the regulator exercises selective enforcement, and that the selection rule is sensitive to the regulated agents' behavior in other domains. When firms that contest regulatory decisions face higher enforcement probability than firms that do not contest, the enforcement differential creates a lateral connection between the domain of the unenforceable norm and any domain in which the firm might otherwise contest.

The third condition is a politically active regulated population. Curvature is visible only when regulated agents have interests that extend beyond the domain of the unenforceable norm. If firms regulated by the environmental norm had no interests in labor policy, trade policy, or fiscal matters, the lateral silencing effect would have no behavioral consequence. Curvature in normative space is observable through its effects on behavior. A politically inert regulated population would produce no observable lateral silencing, even in a system with high structural unenforceability and highly discretionary enforcement. The curvature exists, but it has no observable behavioral signature.

III.2 A Taxonomy of Curvature Types

Not all curvature is alike. Three distinct types can be distinguished, corresponding to the three failure modes of the LSE mechanism and to three different geometric structures.

Type 1 curvature: Lateral convergence. This is the curvature produced by the lateral silencing mechanism in its most developed form. Effects from one regulatory domain converge on behavior in other domains. The parallels meet. In Riemannian geometry, which describes surfaces of constant positive curvature, all initially parallel geodesics converge. The sphere is the canonical example: two lines drawn perpendicular to the equator, initially parallel, meet at the poles. Type 1 normative curvature produces an analogous convergence: enforcement actions in domain A converge on behavior in domains B, C, and D. The normative parallels, the assumption that domain A norms do not affect domain B behavior, fail.

Type 2 curvature: Causal inversion. This is the curvature produced when effects from lower levels in the normative hierarchy propagate upward, against the direction of validity. In standard legal theory, lower norms cannot affect higher norms. In curved normative spaces, the silencing of regulated agents removes a feedback mechanism that would otherwise allow lower-level dysfunction to generate reform pressure on higher-level norms. The Argentine dual-control environmental regime, maintained intact by four governments whose policy preferences differed substantially, illustrates this type: the lower-level regulation (the incompatible Resolution) suppresses the political pressure that would otherwise have generated statutory reform at a higher level. The causal flow runs upward against the validity hierarchy.

Type 3 curvature: Functional superposition. This is the curvature produced when a norm exists in two functional states simultaneously. In standard legal theory, a norm has a determinate function: the function it was enacted to serve. In curved normative spaces, a structurally unenforceable norm exists simultaneously as a regulatory norm (its declared function) and as a silencing instrument (its secondary function). The two functional states are not successive: the norm does not transition from one to the other. It exists in both simultaneously, and the act of enforcement collapses the superposition: each specific enforcement action instantiates one or the other function, but the norm as a structural feature of the legal system retains both.

These three types of curvature are analytically distinct but empirically interrelated. In developed cases of LSE, all three are present simultaneously. The lateral convergence produces the silencing effect; the causal inversion sustains the unenforceable norm by suppressing reform pressure; the functional superposition is what makes both the regulatory and the silencing function simultaneously available to the regulator. The three types are the three faces of the same curvature.

IV. Three Non-Euclidean Geometries for Legal Analysis

IV.1 Riemannian Geometry: The Sphere of High-LSE Systems

Riemannian geometry, named for the same Riemann whose 1854 lecture opened this paper's argument, describes spaces of positive curvature. The canonical Riemannian surface is the sphere. On a sphere, the sum of the angles of a triangle exceeds 180 degrees. Parallel lines, defined as lines that are equidistant at all points, do not exist: any two lines drawn on a sphere will eventually intersect. The shortest path between two points is not a straight line in the Euclidean sense but a great circle arc.

High-LSE normative systems exhibit Riemannian geometry in a specific sense. The parallel tracks of distinct regulatory domains, which in Euclidean normative space run indefinitely without intersection, curve toward each other and converge at the silencing pole. That pole is not a geographic location but a behavioral attractor: the point at which regulated agents have reduced their contestation activity across all domains to its minimum. All regulatory geodesics, the paths along which regulatory effects travel in the normative space, converge at that attractor.

The Riemannian analogy has a further implication. On a sphere, the interior angles of triangles are larger than on a flat surface, and the sum increases with the size of the triangle. The normative analogue: in high-LSE systems, the effective normative scope of each enforcement action is larger than it would be in a flat system, because each enforcement action affects behavior not only in the domain being enforced but in all connected domains. A small enforcement action in the environmental domain has a large normative footprint across the labor, trade, and fiscal domains. The amplification increases with the size of the silencing surface, measured by the number of domains in which the regulated population has interests.

The Constitutional Lock-in Index (CLI) is, in this framework, a measure of the radius of curvature of the Riemannian normative surface. A high CLI system (Argentina, CLI = 0.89) has a small radius of curvature: the surface is tightly curved, the parallels converge quickly, and the silencing pole is relatively near. A low CLI system (Chile, CLI = 0.24) has a large radius of curvature: the surface approaches flat, the parallels converge slowly if at all, and the silencing pole is distant. In the limit, as CLI approaches zero, Riemannian normative geometry approaches Euclidean normative geometry, and the standard tools of legal theory recover their accuracy.

The Riemannian model also illuminates why reform attempts in high-LSE systems tend to fail in a specific way. On a sphere, if you try to draw a straight line, you end up tracing a great circle arc: the curvature of the space bends your path regardless of your intention. In high-LSE normative systems, reform efforts that are designed for flat spaces produce curved results. A legislator who introduces a bill to reform the unenforceable norm is operating in a space where the political support for reform has been systematically suppressed by the silencing mechanism. The reform effort curves back toward the status quo not because of any specific political opposition but because the geometry of the space in which it operates bends it there.

IV.2 Hyperbolic Geometry: The Space of High-Modularity Systems

Hyperbolic geometry describes spaces of constant negative curvature. The canonical hyperbolic surface is the saddle or, more precisely, the Poincare disk, a model in which the entire infinite hyperbolic plane is represented inside a unit disk, with distances expanding exponentially toward the boundary. On a hyperbolic surface, the sum of the angles of a triangle is less than 180 degrees. Through any point not on a given line, there are infinitely many parallel lines, all of which fail to intersect the original. The parallels diverge instead of converging.

Hyperbolic normative geometry describes a different limiting case: systems of high modularity, where the distinct regulatory domains are insulated from each other by design. In such systems, an enforcement action in domain A has effects that decay rapidly in domains B, C, and D. The parallels diverge: actions in one domain become asymptotically irrelevant to behavior in other domains as the domains become more distant in the normative space.

The Institutional Evolvability Index (IEI) captures this dimension of normative geometry. The modularity component (M) of the IEI measures precisely the capacity to reform subsystems

without fragmenting the whole: in geometric terms, it measures the degree to which distinct domains are separated by normative curvature of the hyperbolic type. A high-M system has domains that are genuinely independent, where actions in one do not curve into others. A low-M system has domains that are tightly coupled, where actions in one propagate into all others, corresponding either to high positive curvature (Riemannian, if the propagation is silencing) or to a degenerate flat system where domains are formally separate but substantively linked.

The normative implications of hyperbolic geometry are less obviously pathological than those of Riemannian geometry, but they are not neutral. A hyperbolic normative space, while insulating domains from each other, also makes coordination across domains more difficult. If the environmental domain and the labor domain diverge hyperbolically, a regulated agent who wants to engage on both simultaneously faces a normative space in which the two domains are increasingly distant: the resources invested in one domain produce diminishing returns in the other. This describes a specific pattern of regulatory engagement: specialization of contestation, where firms and associations become expert in one regulatory domain and abstain from others not because they are silenced but because the hyperbolic geometry makes cross-domain engagement inefficient.

The distinction between Riemannian and hyperbolic curvature at the behavioral level is the following. In Riemannian high-LSE systems, regulated agents are silenced across all domains simultaneously, because the convergence of parallels means that enforcement in any one domain raises the threat in all others. In hyperbolic high-modularity systems, regulated agents specialize in a single domain and abandon others, because the divergence of parallels means that engagement in one domain has no spillover benefits in others. Both produce lateral suppression of contestation in all but one domain. The mechanism is different: coercion in the Riemannian case, rational specialization in the hyperbolic case. An empirical legal researcher who observes reduced contestation across domains without distinguishing these mechanisms will misdiagnose the cause and prescribe the wrong remedy.

IV.3 Quantum Superposition: The Indeterminate Norm

The quantum mechanics analogy requires more careful handling than the geometric analogies, because quantum mechanics is a physical theory with a specific mathematical structure, and importing it wholesale into legal theory risks producing confusion rather than insight. The claim here is limited and precise: certain structural features of quantum mechanical systems, specifically the superposition of states and the collapse of the wave function through measurement, correspond to structural features of curved normative systems in ways that illuminate phenomena that standard legal theory cannot represent.

In quantum mechanics, a particle can exist in a superposition of two or more states simultaneously. The superposition is not a classical mixture, where the particle is in one state or the other with some probability: it is a genuinely indeterminate condition in which both states are

simultaneously real. The act of measurement collapses the superposition: after measurement, the particle is in one definite state. The measurement itself, not the pre-existing state of the particle, determines the outcome.

Structurally unenforceable norms exhibit an analogous structure. Such a norm exists simultaneously in two functional states: the regulatory state, in which it operates as a rule requiring conduct; and the silencing state, in which it operates as an instrument of political control. These two states are not successive: the norm does not transition from regulatory to silencing function at some identifiable point. Both functions are simultaneously real, available to the regulator in each specific enforcement action. The enforcement act collapses the superposition: a specific enforcement action instantiates either the regulatory function (when it is directed at a firm that is genuinely out of compliance on substantive grounds) or the silencing function (when it is directed at a firm that has contested a regulatory decision). But the norm as a structural feature of the legal system retains both functional states between enforcement actions.

This has implications for the concept of legal interpretation. Standard interpretive theory asks: what does this norm mean? The question presupposes that the norm has a determinate meaning that the interpretation aims to uncover. In the quantum superposition model, the question is malformed: the norm does not have a single determinate functional state until the enforcement act collapses the superposition. The interpretive question should be: in what functional state will this norm be collapsed by the next enforcement act, and what are the conditions that determine the collapse? That is a different question, and it requires different analytical tools.

The quantum analogy also illuminates the observer effect in legal interpretation. In quantum mechanics, the act of measurement disturbs the system being measured. In curved normative spaces, the act of legal interpretation, the public articulation of what a norm means and how it should be applied, disturbs the normative system by making the silencing function more or less visible. A judicial opinion that identifies the silencing function of a structurally unenforceable norm collapses the superposition in the regulatory direction, by making it harder for the regulator to use the norm as a silencing instrument without this use being publicly identified as such. Interpretive activity in curved normative spaces is not merely descriptive: it participates in determining the functional state of norms whose states are otherwise indeterminate.

V. Diagnostic Instruments as Curvature Measures

V.1 The CLI as Radius of Curvature

The Constitutional Lock-in Index measures institutional rigidity as a function of four components: precedent density (P), doctrinal elaboration (D), professional organization (O), and enforcement infrastructure (E), combined according to the formula $CLI = 0.30P + 0.25D + 0.20O + 0.25E$. In geometric terms, the CLI is a measure of the radius of curvature of the normative surface. A high CLI corresponds to a small radius of curvature, meaning a tightly curved surface

on which the Euclidean parallels converge rapidly. A low CLI corresponds to a large radius of curvature, meaning a nearly flat surface on which the parallels remain approximately parallel over long normative distances.

The empirical calibration of the CLI against reform outcomes supports this interpretation. The framework predicts that reforms in high-CLI systems fail not because of specific political opposition but because the geometry of the normative space bends them back toward the status quo: CLI > 0.70 is associated with a 95% failure rate for structural reforms across 60 cases in four jurisdictions. In Riemannian geometry, there is a direct relationship between curvature and the convergence rate of initially parallel geodesics: higher curvature means faster convergence. The CLI-to-reform-failure relationship is the empirical expression of that geometric relationship in normative space.

The CLI also determines the range of the Riemannian curvature. On a sphere of radius r , the maximum distance between two points is π times r : points that are further apart than that are actually getting closer to each other, because they have passed the equator and are approaching the opposite pole. In normative terms: in a high-CLI system, a regulatory action that would, in a flat system, push the normative state away from the silencing equilibrium may, in the curved system, bring it closer. Reform efforts that appear to be moving away from the dysfunctional equilibrium may be following a Riemannian geodesic that curves back toward it. The reform paradox documented in the CLI validation data, where crisis reinforces lock-in in high-CLI systems with a statistically significant interaction term, is the normative analogue of this Riemannian property.

V.2 The IHR as Accumulated Curvature

The Institutional Hysteresis Rate measures the accumulated deformation of the institutional environment: the residual distortion that persists after a norm has been formally repealed or modified. In geometric terms, the IHR measures the curvature that has been permanently imparted to the normative surface by prior regulatory actions, a curvature that persists after the specific norms that caused it have been removed.

The hysteresis concept from physics is precisely applicable here. In materials science, hysteresis describes the path-dependence of a material's response to external forces: the deformation caused by applying a force is not fully reversed when the force is removed, because the material's internal structure has been permanently altered. The normative analogue is the infrastructure built around a norm, the specialized courts, bar associations, academic doctrine, enforcement agencies, that persists after the norm itself has changed. That infrastructure is not simply a residue; it is a source of continued curvature in the normative space, bending future regulatory actions toward the forms that the pre-existing infrastructure supports.

In the Argentine labor law context, the IHR for the Article 14bis framework is estimated at approximately 0.87. This means that 87% of the institutional deformation caused by that framework's regulatory regime persists independently of the formal validity of the specific norms.

In geometric terms: the curvature imparted to the normative space by Article 14bis and its associated enforcement infrastructure is 87% permanent. A legislator who repeals Article 14bis tomorrow would be operating in a normative space whose curvature has been reduced by only 13%. The remaining 87% of curvature would continue to bend regulatory actions, interpretive choices, and institutional behavior toward the forms that the pre-existing infrastructure supports.

V.3 The LSI as Active Curvature

The Lateral Silencing Index, proposed in Lerer (2026b) and developed operationally as a four-component instrument, measures the active curvature being produced in a normative space at a given moment. While the CLI measures the structural rigidity of the surface and the IHR measures the accumulated deformation, the LSI measures the rate at which new curvature is being produced by ongoing silencing mechanisms.

The four components of the LSI correspond to four dimensions of the curvature-production process. The Unenforceability Score (U) measures the proportion of regulated agents that cannot achieve simultaneous compliance: this is the structural precondition for curvature, corresponding to the violation of Fuller's eighth principle. The Enforcement Differential (ED) measures the ratio of enforcement actions against contesting firms to enforcement actions against non-contesting firms: this is the mechanism through which the curvature is translated into behavioral effects. The Domain Spread (DS) measures the number of regulatory domains in which the lateral suppression of contestation is observable: this is the width of the curvature, corresponding to the angular excess of the Riemannian triangle. The Persistence Indicator (PI) measures the number of government cycles across which the unenforceable norm has remained unreformed: this is the temporal depth of the curvature, indicating how long the normative space has been curved in this way.

Together, CLI, IHR, and LSI constitute a three-dimensional curvature tensor for normative space. The CLI is the structural curvature, the curvature that would exist in the normative space even in the absence of active silencing. The IHR is the historical curvature, the curvature imparted by prior regulatory activity that persists regardless of current norm content. The LSI is the active curvature, the curvature being produced by the current operation of the silencing mechanism. A complete geometric diagnosis of a normative system requires all three.

VI. Theoretical Implications

VI.1 Implications for Legal Philosophy

The non-Euclidean framework developed in this paper has implications for several long-standing debates in legal philosophy. The most immediate concerns the relationship between legal validity and legal efficacy.

Kelsen's separation of validity from efficacy, discussed in Section II.1, was designed to prevent the theory of law from collapsing into sociology: the validity of a norm should not be hostage to the contingent fact of whether it is obeyed. This is a principled position, and the present framework does not challenge it directly. But it does show that the separation has a geometric cost: by rendering efficacy normatively invisible, the pure theory cannot represent the curvature that efficacy failures produce in normative space. A theory of legal validity that is adequate for flat normative spaces may be systematically misleading when applied to curved ones.

The implication is not that Kelsen was wrong about validity but that the pure theory's sphere of application is smaller than its proponents have assumed. It is the geometry of flat normative space. In curved spaces, the tools it provides are insufficient for the analytic tasks that legal theorists and legal practitioners need to perform. The relationship between validity and efficacy in curved spaces is not simple: a valid but unenforceable norm can be the primary driver of normative curvature, producing effects that a purely validity-focused analysis would miss entirely.

Hart's account of secondary rules faces an analogous limitation. The rule of recognition, as a social practice among officials, is designed to identify which norms are valid. In flat normative space, that identification exhausts the normative work: once a norm is identified as valid, its position in the hierarchy is determined, and the analysis can proceed. In curved normative space, identifying a norm as valid leaves open the most important question: in what functional state will the norm be when it is applied, and how will that application affect the curvature of the space in which subsequent regulatory actions take place? The rule of recognition is an instrument calibrated for flat space. It measures the wrong dimension when the space is curved.

Fuller's contribution is partially salvaged by the present framework, but in a way he would not have anticipated. His eight principles of legality are not the conditions for law's existence, as he believed, but the conditions for flat normative space. When they are violated, law does not cease to exist: the normative space curves. Fuller saw the violation as pathology; the present framework sees it as a geometric transformation. The normative space that emerges from the violation of Fuller's principles is not less real than the flat space produced by their observance; it is differently shaped, with different properties and different implications for the behavior of regulated agents.

VI.2 Implications for Institutional Design

The geometric framework has direct implications for institutional design, specifically for the design of regulatory systems and the conditions under which reform efforts can succeed.

The most immediate implication is that regulatory design must account for curvature. A regulator who designs a norm for a flat normative space, without attending to the curvature that the norm will produce or the existing curvature in which it will operate, is in the position of an architect who designs a building for a flat surface and then places it on a hillside. The building

may be internally consistent, but its relationship to its environment will produce structural stresses that the design did not anticipate.

Specifically: in high-curvature environments, the secondary effects of norms will dominate their primary regulatory effects. A norm that produces high LSE will suppress the political feedback that would otherwise allow it to be reformed when its primary function fails. The norm becomes self-sealing: its failure to achieve its declared purpose removes the political actors who would otherwise demand its reform. Institutional designers who do not model this dynamic will be surprised by the persistence of failing norms in high-curvature environments.

The pre-deployment normative evaluation framework developed in Lerer (2026d; DOI 10.5281/zenodo.18947186) provides a partial response to this problem. By requiring assessment of a norm's potential side effects before it is enacted, the framework creates an opportunity to identify high-LSE potential before the curvature is produced. The present paper provides the theoretical foundation for that requirement: high-LSE potential corresponds to high curvature potential, and high curvature potential means that the norm's secondary effects will dominate its primary function once the enforcement differential becomes established.

A second institutional design implication concerns the sequencing of reforms in high-curvature environments. In Riemannian normative space, as noted in Section IV.1, a reform effort that is designed for flat space may follow a geodesic that curves back toward the dysfunctional equilibrium. The implication is that successful reform in high-curvature environments requires first reducing the curvature and then introducing the substantive reform. In practice, this means addressing the enforcement differential (by reducing the asymmetry between the treatment of contesting and non-contesting firms) before or simultaneously with addressing the structural unenforceability (by reforming the incompatible norms). The Saakashvili reforms in Georgia provide the most documented example: the January 2004 reforms addressed structural unenforceability and enforcement architecture simultaneously, which had the effect of both eliminating the source of curvature and removing the existing curvature in a single legislative action.

VI.3 Implications for Empirical Legal Studies

The geometric framework also has implications for empirical legal research. The central implication is that studies of regulatory compliance, regulatory effectiveness, and legal reform that assume flat normative space will produce systematically biased results in curved environments.

Consider regulatory compliance studies. Standard compliance research measures the gap between what a norm requires and what regulated agents do. In flat normative space, this gap is a straightforward function of enforcement probability, penalty magnitude, and the costs of compliance. In curved normative space, the gap has an additional component: the contribution of the enforcement differential, which creates an incentive to remain nominally non-compliant because compliance would remove the vulnerability that the silencing mechanism requires. A

regulated agent in a high-LSE system who fully complied with the unenforceable norm would eliminate the regulator's leverage over them. Full compliance would, paradoxically, increase the agent's political exposure by eliminating the pretext for discretionary enforcement in retaliation for political contestation. The compliance study that does not model this incentive will misidentify the source of non-compliance and prescribe remedies that address the wrong mechanism.

Regulatory effectiveness studies face a related problem. Standard effectiveness research asks whether a norm achieves its declared purpose. In flat normative space, this is the right question. In curved normative space, the declared purpose may be irrelevant to the norm's actual function, which is lateral silencing. A norm that achieves zero compliance with its declared environmental standards while producing complete suppression of political contestation across all regulatory domains is, by the standard metric, a complete failure. By the metric appropriate to curved normative space, it is a complete success, in the sense that it is achieving its actual functional purpose with high efficiency. The effectiveness study that asks only about declared purposes will misclassify this outcome.

VII. Falsifiable Predictions

The non-Euclidean normative space framework generates predictions that are distinct from those of standard legal theory and testable with available data. Five predictions follow directly from the geometric analysis.

Prediction 1 (Curvature-reform correlation).

Systems with higher CLI scores should show not only higher rates of reform failure but a specific pattern of failure: reforms that begin as apparent successes before curving back toward the pre-reform equilibrium. This is the Riemannian geodesic prediction. In flat systems, a reform that achieves initial adoption should either succeed or fail to be implemented, with no systematic tendency to curve back. In Riemannian systems, initial adoption followed by gradual erosion should be the modal pattern. The Argentine labor reform attempts between 1991 and 2024 provide a partial test dataset; the full test would require comparable longitudinal data from other high-CLI jurisdictions.

Prediction 2 (Observer effect in interpretation).

Judicial or administrative decisions that explicitly identify the silencing function of a structurally unenforceable norm should produce measurable reductions in the enforcement differential in the period following the decision. The interpretive act collapses the superposition in the regulatory direction by making the silencing function publicly visible and therefore more costly for the regulator to exercise without challenge. This prediction is falsified if explicit judicial identification of the silencing function produces no change in enforcement differential patterns.

Prediction 3 (Curvature-specialization correlation).

In hyperbolic high-modularity systems, regulated entities should show a pattern of regulatory specialization rather than lateral suppression: high engagement in one regulatory domain combined with low engagement in others, driven by efficiency rather than coercion. This is the hyperbolic geometry prediction. It is distinguished from the Riemannian prediction by the mechanism: in Riemannian systems, lateral suppression is correlated with enforcement events; in hyperbolic systems, specialization is uncorrelated with enforcement events. The two predictions can be empirically distinguished by examining whether cross-domain engagement reduction follows enforcement actions or precedes them.

Prediction 4 (CLI-LSI interaction).

The relationship between the CLI and the LSI should be non-linear and asymmetric. At low CLI values, increases in LSI should have small effects on reform probability, because the flat or nearly flat normative space does not amplify the curvature produced by silencing. At high CLI values, increases in LSI should have large effects on reform probability, because the existing structural curvature amplifies the additional curvature produced by silencing. The interaction term CLI x LSI should be statistically significant and positive in a regression of reform failure probability on CLI, LSI, and their interaction.

Prediction 5 (Simultaneous reform sequencing).

Reform attempts in high-curvature environments that address structural unenforceability and enforcement architecture simultaneously should have higher success rates than reform attempts that address only one dimension. This is the institutional design prediction derived from the Riemannian geometry analysis. Reforms that eliminate the incompatible norm but leave the enforcement differential intact should fail at higher rates than simultaneous reforms, because the remaining enforcement differential continues to curve the normative space toward the silencing equilibrium even after the formal source of structural unenforceability has been removed.

VIII. Epistemological Positioning

This paper is a contribution to a Lakatosian research program (Lakatos, 1978) whose hard core consists of three commitments: that legal norms are cultural replicators subject to evolutionary selection (Dawkins, 1976; Hull, 1988); that legal institutions are extended phenotypes of those replicators (Dawkins, 1982); and that evolutionary game theory provides the appropriate formalism for modeling institutional equilibria (Maynard Smith, 1982). The non-Euclidean normative space framework operates in the protective belt of that program: it introduces a new conceptual instrument, the geometry of normative space, that is consistent with the hard core and extends the program's explanatory reach to phenomena, specifically, the systematic failure of

standard legal-theoretic tools in curved normative environments, that prior components of the program did not address.

The paper's relationship to the external literature on legal theory is one of extension and reframing rather than refutation. Kelsen, Hart, Fuller, and Raz are not wrong on their own terms. They are right about flat normative space, and flat normative space is a good approximation for many legal systems and many regulatory contexts. The non-Euclidean framework does not invalidate their work; it identifies the domain within which that work is accurate and describes what happens outside that domain. This is precisely the relationship between Euclidean and non-Euclidean geometry: Euclid is not wrong, he is right about flat surfaces. The present paper argues that normative surfaces are not always flat, and that a complete theory of law must account for their curvature.

The epistemological stance follows Popper (1959) on demarcation: the paper provides falsifiable predictions in Section VII. It follows Campbell (1974) on evolutionary epistemology: the framework is offered as a hypothesis subject to empirical selection, not as a deductive derivation from first principles. The five predictions are stated in a form that allows clear falsification, and the paper is committed to the view that the framework should be abandoned or substantially revised if the predictions are consistently disconfirmed.

IX. Limitations

Four limitations merit explicit acknowledgment before the conclusion.

First, *the geometric analogies employed in this paper are heuristic rather than mathematical*. The paper does not develop a formal metric for normative space or derive the predictions from a rigorous geometric apparatus. It uses the conceptual structure of non-Euclidean geometry, the failure of the parallel postulate, the distinction between positive and negative curvature, the collapse of superposition, as organizing frameworks for thinking about properties of normative systems that standard legal theory cannot represent. The value of those frameworks is heuristic: they organize observations and generate predictions in a way that the flat-space frameworks of standard legal theory do not. A full mathematical development of the framework would require formal definitions of normative distance, normative geodesics, and normative curvature that go well beyond what this paper provides.

Second, *the empirical evidence for the specific claims about curvature in the Argentine and Georgian cases rests on the case studies developed in Lerer (2026b)*. That paper identified the structural unenforceability of the relevant norms, the discretionary enforcement patterns, and the lateral suppression of contestation as observable phenomena in those cases. The present paper reinterprets those observations in geometric terms. The reinterpretation does not generate new empirical claims; it provides a framework for understanding the significance of the existing observations. The empirical limitations of the case studies, including the unverified source issues

identified in the revision process for Lerer (2026b), apply equally to the geometric interpretation offered here.

Third, *the quantum superposition analogy requires more careful handling than the paper has given it*. The claim that structurally unenforceable norms exist in a superposition of functional states is not a claim about quantum mechanical properties of legal systems; it is a claim about the indeterminacy of functional states in a class of normative phenomena. The analogy illuminates that indeterminacy and suggests a specific mechanism, the collapse through enforcement action, by which it is resolved. But the analogy should not be taken to imply that the mathematics of quantum mechanics applies to normative systems, or that the specific formal apparatus of quantum field theory can be imported into legal theory without substantial modification.

Fourth, *the framework developed here focuses exclusively on the curvature produced by the LSE mechanism*. There may be other sources of normative curvature that are not captured by the LSE framework: international law's relationship to domestic law, the effects of constitutional adjudication on the structure of lower-level norms, the curvature produced by informal norms that operate alongside formal legal norms. These are important topics that the present framework opens up but does not address. A complete theory of normative geometry would need to account for all sources of curvature, not only those identified by the LSE mechanism.

X. Conclusions

Legal theory has operated, without examination, on the assumption that normative space is flat. That assumption is adequate for simple, well-functioning legal systems in which norms are structurally enforceable, enforcement is systematic, and secondary effects are negligible. It is not adequate for the large and practically important class of normative systems in which structurally unenforceable norms produce lateral silencing effects, enforcement differentials bend regulatory behavior across domains, and the declared functions of norms diverge systematically from their actual functions.

In those systems, normative space curves. The curvature is not a metaphor: it corresponds to specific, identifiable features of regulatory environments that produce predictable deviations from the behavior that flat-space legal theory would expect. The Euclidean parallels meet. The sum of the angles of the normative triangle exceeds or falls short of 180 degrees. The wave function of the norm collapses in the enforcement act rather than being determined in advance by the norm's declared content.

The non-Euclidean framework developed in this paper is a first step toward a geometry of normative space adequate to these phenomena. It connects three existing bodies of work, the EPT research program, the mathematics of non-Euclidean geometry, and the analytic tradition in legal philosophy, in a way that none of them has previously been connected. It generates testable predictions that distinguish it from both the flat-space frameworks of standard legal theory and the

informal social-science accounts of regulatory dysfunction. And it provides a theoretical foundation for the pre-deployment normative evaluation framework and the Lateral Silencing Index that have been proposed elsewhere in this research program.

Legal theory has not yet had its Riemann. Whether the framework developed here proves to be the right geometry for normative space, or merely a step toward it, the question it opens, what is the geometry of the space in which law operates, is one that the discipline can no longer avoid. The flat space assumption has been adequate for long enough. The curved cases are too important, and too common, to continue treating as anomalies.

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